



ENHANCING SKILLS AND OPPORTUNITIES WITH IBM ZXPLORE

MOLDOVAN LUCA

ABOUT ME

Who am I?

My name is Moldovan Luca, I am a 19-year-old Computer Science student minoring in Electrical Engineering at Constructor University in Bremen. Originally from Romania, I am honored to serve as a member of the leading board of the IBM Z Club at our university.

IBM Z AT CONSTRUCTOR UNIVERSITY

What is our purpose?

The IBM Z Club at Constructor University is dedicated to supporting our members in earning IBM's professional badges, which we will discuss further. These badges not only enhance technical expertise but also open doors to exciting opportunities. By achieving these certifications, we aim to encourage our members to participate in virtual career fairs, enabling them to connect with industry professionals and advance their careers.

IBM Z XPLORE LEARNING PLATFORM

What is IBM Z Xplore?

IBM Z Xplore is a comprehensive and fun, challenge-based digital platform designed to facilitate learning, exploration, and evaluation of IBM Z technology. Available globally at no cost and requiring no prior knowledge, it provides access to IBM Z cloud infrastructure, tools, code samples, tutorials, and more. The platform features hands-on labs and challenges, offering learners the opportunity to gain in-demand IBM Z skills through practical experience, supporting career growth in enterprise computing which powers 68% of all worldwide transactions.

IBM BADGES

In today's session, we will explore two key IBM badges: "Fundamentals and Concepts" and "Advanced," highlighting their significance and value in skill development. These badges are awarded to learners upon successfully completing a series of challenges. In this session, we will delve into the details of these challenges and the valuable skills acquired through earning these two industry recognised digital badges.



FUNDAMENTALS + CONCEPTS BADGE

The IBM Z Concepts Badge represents the acquisition of key skills essential for effectively managing tasks in an IBM Z mainframe environment. These skills are critical for ensuring the smooth operation, efficiency, and optimization of mainframe systems, which are integral to many industries requiring high-performance computing.

Key skills and knowledge covered:

1. **Z Systems Navigation and Data Management** – ability to effectively navigate through the mainframe's Zowe interface and USS (Unix System Services) environment. Learners will have an understanding of **Data Set Filters, Partitioned Data Sets (PDS), Members within a PDS.**
2. **JCL (Job Control Language)** - JCL is a cornerstone of mainframe operation, and mastery of its syntax is vital for automating batch jobs, an essential daily task. **JCL Structure and Syntax, Optimized Data Management.**
3. **Unix System Services (USS) and Command Line Skills** - Proficiency in USS scripting and command-line operations is crucial for managing tasks in a Unix-like environment on the IBM Z mainframe. **System Task Management, Linux/Unix Familiarity.**

MORE ABOUT THE CONCEPTS BADGE

Key skills and knowledge covered:

1. **REXX Programming Language** - REXX is widely used for automation and scripting in IBM Z environments, especially in z/OS. **Creating and Running Scripts, Programming Structures, Library Functions.**
2. **Python Programming and Integration** - Python is increasingly used for scripting and automation in the IBM Z environment due to its versatility and simplicity. **Advanced Utilities, Advanced Integration**

Summary of the skills gained – Bash (Unix shell and command language), **JCL, REXX, Programming, Python, Zowe** (open source software framework that provides solutions that allow development and operations teams to securely, manage, control, script and develop on the Mainframe), **z/OS** (64-bit operating system for IBM z/Architecture mainframes), **Unix System Services(USS).**

ADVANCED BADGE

This badge recognizes advanced mainframe expertise, highlighting the ability to solve complex problems using critical thinking and diverse resources. The earner demonstrates proficiency in system orchestration with Ansible and Zowe, working in Docker containerized environments, and programming in REXX, COBOL, Linux, and JCL. Their strong conceptual understanding and technical skills make them well-equipped for modern enterprise IT challenges.

New key skills gained:

1. **Multi OS Proficiency** - The badge emphasizes the ability to seamlessly manage multiple operating systems on IBM Z, such as z/OS and Linux, essential for hybrid IT environments. **z/OS for enterprise computing, Linux integration.**
2. **Data Management and Relational Databases** - Proficiency in managing complex datasets and databases is a cornerstone of IBM Z expertise. **z/OS Datasets, SQL and Db2.**
3. **Programming for Enterprise Systems** - Advanced skills in languages tailored to IBM Z's robust environment prepare professionals to maintain and develop mission-critical applications. **COBOL, Assembler Programming.**

MORE ABOUT THE ADVANCED BADGE

More key skills gained:

1. **Automation with JCL, REXX and Python** - Automating tasks is vital for efficiency in enterprise environments, and this badge validates expertise in key automation tools. **JCL (Job Control Language), REXX Scripting, Python.**
2. **Zowe CLI and Unix System Services** - Practical experience with Zowe CLI and USS equips professionals with modern tools to manage mainframe operations. **Zowe CLI, Unix Commands.**

Summary of some of the skills gained - **Ansible** (open-source tool for provisioning, configuration management, and application deployment, supporting infrastructure as code), **COBOL, Db2, Docker** (PaaS solution that uses OS-level virtualization to deliver software via containers), **REXX, REGEX, TSO, VSAM, Linux.**

IBM Z PERSONAS

System Administrator Modernization Architect Application Developer

Mainframe System Administrator

This learning track includes paths to gain the knowledge that is needed as a Mainframe System Administrator. These tracks contain over 300 hours of self-paced, hands-on training with foundational and advanced topics. Skills include z/OS system administration, CICS, Db2, IMS and more. Earn industry-recognized Credly digital badges to elevate your skills to employers as you begin your career.



System Administrator **Modernization Architect** Application Developer

Modernization Architect

This learning track explores the seven modernization patterns to architect solutions to meet your modernization business objectives. The result is an agile and seamlessly integrated hybrid cloud platform with a modern mainframe at the core. Skills include Db2, z/OS Connect, Automatic Binary Optimizer (ABO), Open Data Analytics, AI, COBOL, and more.



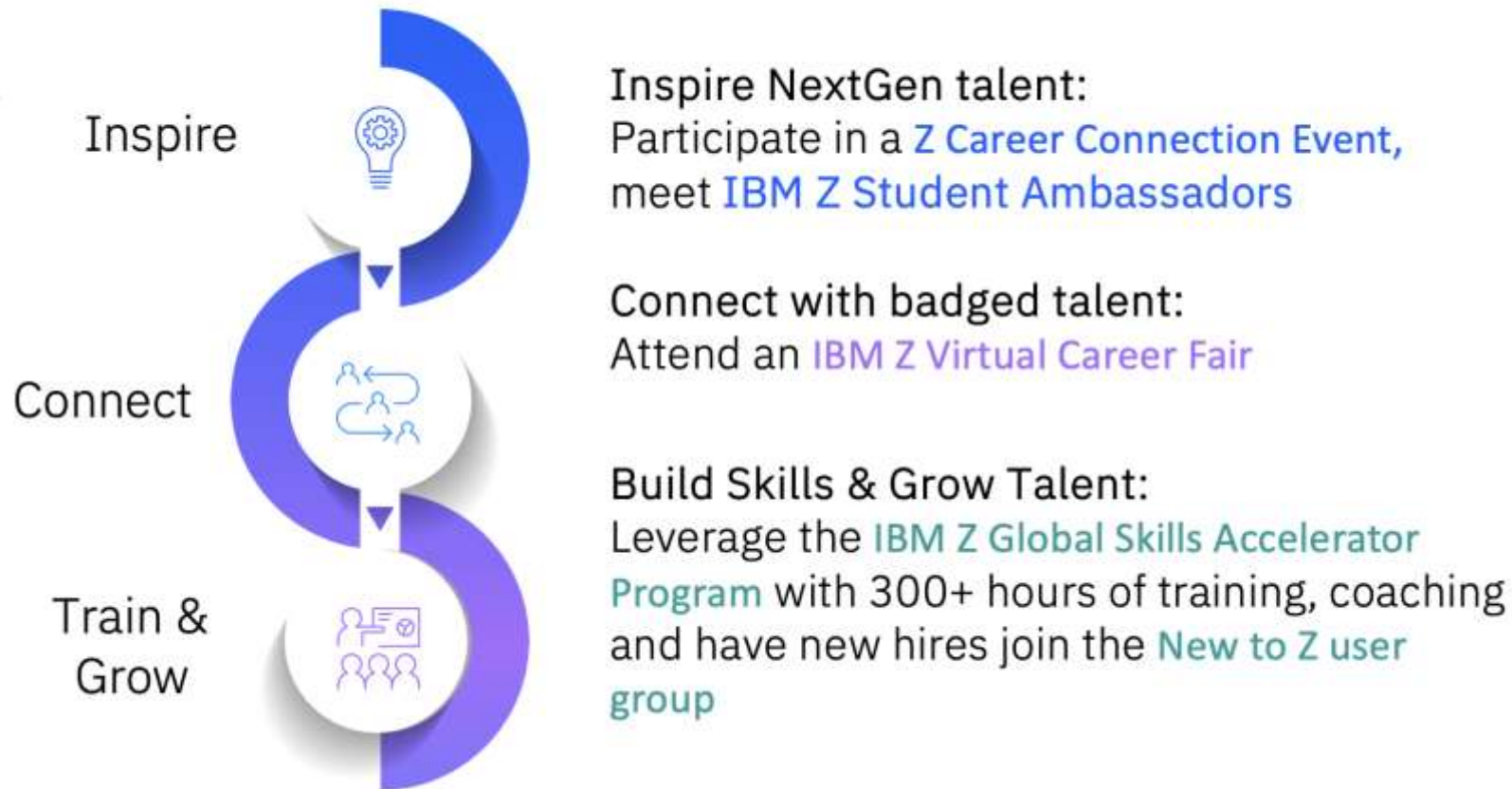
System Administrator Modernization Architect **Application Developer**

Mainframe Application Developer

This learning path tracks over 250 hours of self-paced hands-on training to prepare you for the Mainframe Application Developer. Discover Agile Practices, Big Data and Analytics, Web Development, Programming Languages, DevOps and IBM Applications. Skills include COBOL, Java, Python, CICS, IMS, GitHub and more.



IBM Z PROGRAMS



INSPIRE NEXT-GEN TALENT

IBM Z Career Connection

Meet and inspire local students to explore mainframe technologies and careers.

116

2024 University
Events

10200

Students Reached



On-Campus IBM Z Student Ambassadors

Engage with students on campuses all over the world who are building and leading Mainframe Student Clubs.

178

On Campus
Ambassadors

118

Universities
20 Countries

IBM Z Student Ambassadors

Powered by Your Big Year

